Questions by topic: Conditions for life on Earth - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 1 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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0 1 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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0 1 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

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0 1 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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0 2 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

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0 2 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

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0 2 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

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0 3 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

0 3 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

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0 3 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores .....

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Gas analysis of ice cores .....

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Radiocarbon dating .....

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Magnesium : calcium ratio in calcite .....

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0 4 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 4 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 4 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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0 5 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 5 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer _____ million years

0 5 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 5 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 5 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 5 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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0 6 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

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0 6 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

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0 6 . 3 Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

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0 6 . 4 Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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END OF QUESTIONS

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